CHAPTER 4

WEB TECHNOLOGY

CHAPTER ESSENCE AT A GLANCE:
Web technology enables globally distributed client-server communication across the internet through standardized request-response protocols, DNS resolution pipelines, and tiered client-server scripting environments.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Client-Server Distributed Computing:** The World Wide Web functions through a distributed model where client user-agents dispatch HTTP requests over network sockets to listening server daemon processes that process resources and return formatted HTTP responses.

• **Static vs Dynamic Web Paradigms:** Static web pages deliver identical pre-stored HTML documents across all requests with minimal server processing, whereas dynamic web pages generate customized content on-the-fly using database queries and runtime scripting.

• **Client-Side vs Server-Side Scripting Execution:** Client-side scripts execute locally within the browser engine for interface responsiveness and form validation, while server-side scripts run on the host server to perform business logic, database queries, and secure transactions.

• **Standard Web Protocols and Port Architecture:** Web transactions depend on standardized protocols mapped to dedicated communication ports, including unencrypted HTTP on port 80, TLS-encrypted HTTPS on port 443, FTP on port 21, and DNS on port 53.

ESSENTIAL ARCHITECTURE & CORE SYNTAX CARDS

$$\text{HTTP Request-Response Cycle} = \frac{\text{Client Browser} \overset{\text{GET or POST Request}}{\longrightarrow} \text{Web Server (Port 80 or 443)}}{\text{Client Browser} \overset{\text{Status Code} + \text{HTML or JSON}}{\longleftarrow} \text{Web Server}}$$

HTTP Transaction Architecture

$$\text{Scripting Tier Architecture} = \frac{\text{Client-Side: JavaScript, HTML, CSS (Browser Execution)}}{\text{Server-Side: PHP, Python, Node.js (Server Database Access)}}$$

Two-Tier Web Execution Environment

$$\text{DNS Resolution Pipeline} = \text{URL} \overset{\text{Local Cache}}{\longrightarrow} \text{Root Server} \longrightarrow \text{TLD Server (.org)} \longrightarrow \text{Authoritative Server} \rightarrow \text{IP}$$

Hierarchical DNS Lookup Process

$$\text{Web Protocol Port Mapping} = \text{HTTP}:80, \quad \text{HTTPS}:443, \quad \text{FTP}:21, \quad \text{DNS}:53, \quad \text{SMTP}:25$$

Standard Internet Application Ports

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Never confuse client-side scripting with server-side scripting; client-side scripts run inside the browser and cannot directly access server databases or sensitive credentials.

• **Alert:** HTTP transmits all packets in cleartext plain text; HTTPS uses Transport Layer Security (TLS and SSL) encryption over port 443 to secure passwords and financial transactions.

• **Alert:** A domain name is an alphanumeric mnemonic label resolved by DNS to a 32-bit IPv4 or 128-bit IPv6 numerical address; the domain name itself is not the IP route.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between static web pages and dynamic web pages with two points of comparison. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Static Web Pages: Content remains identical for all visitors, created with simple HTML and CSS, stored pre-built on the server, and loads very rapidly without server execution. **[1 Mark]**

• Dynamic Web Pages: Content is generated at runtime based on user requests, database queries, and server scripts, providing personalized data (e.g. social feeds, shopping carts). **[1 Mark]**

2. Compare client-side scripting and server-side scripting with suitable language examples. [Score: 2 Marks]

Answer & Key Valuation Steps:

• Client-Side Scripting: Executes inside the client web browser to validate forms, animate interfaces, and respond to user clicks without contacting the server (e.g. JavaScript, VBScript). **[1 Mark]**

• Server-Side Scripting: Executes on the web server to query databases, authenticate users, and render HTML before transmission to the client (e.g. PHP, Python, ASP.NET). **[1 Mark]**

3. Explain the step-by-step working of the Domain Name System (DNS) when resolving a web address like www.kerala.gov.in. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Query Initiation and Local Cache: The browser first checks local cache; if unresolved, dispatches a DNS query to the local recursive resolver. **[1 Mark]**

• Hierarchical DNS Lookup: The resolver contacts Root DNS Servers, which direct it to Top-Level Domain (TLD) servers (.in), which in turn route to Second-Level servers (.gov.in). **[1 Mark]**

• Authoritative IP Return: The Authoritative DNS Server for kerala.gov.in returns the exact numeric IP address to the resolver, which caches it and forwards it to the browser. **[1 Mark]**

4. Describe the client-server architecture model of the web with a clear labeled explanation of its communication flow. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Client Role: The web browser acts as the client agent that formats user requests into standardized HTTP requests and sends them across the network. **[1 Mark]**

• Server Role: The web server runs daemon software listening on port 80 or 443, accepts incoming connections, locates or computes the requested resource, and returns an HTTP response. **[1 Mark]**

• Stateless Protocol: HTTP is a stateless protocol where each request-response transaction is handled independently without retaining historical state by default. **[1 Mark]**

5. What is a web server? List three popular web server software packages and describe their primary operational functions. [Score: 3 Marks]

Answer & Key Valuation Steps:

• Definition: A specialized hardware system running software designed to store web documents and serve them to clients across the Internet using HTTP/HTTPS protocols. **[1 Mark]**

• Popular Packages: Apache HTTP Server (open-source industry standard), Nginx (high-performance concurrent server), and Microsoft Internet Information Services (IIS). **[1 Mark]**

• Core Functions: Managing virtual hosts, processing client connections, enforcing access security, serving static assets, and routing requests to server-side scripting runtimes. **[1 Mark]**

6. What are network ports? State the default port numbers for HTTP, HTTPS, FTP, and DNS protocols. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Port Definition: A logical communication endpoint identified by a 16-bit number that directs network data to a specific software service on a host. **[1 Mark]**

• HTTP and HTTPS: HTTP uses port 80 for unencrypted communications; HTTPS uses port 443 for TLS and SSL encrypted secure communications. **[1 Mark]**

• FTP and DNS: FTP uses port 21 for file transfer control commands; DNS uses port 53 for domain name resolution queries. **[1 Mark]**

7. Discuss the technological components required to build and host a modern website, detailing web browsers, web servers, scripting engines, and database backends. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• Web Browser: Client-side rendering engine (Chrome, Firefox, Safari) that interprets HTML, executes JavaScript, and applies CSS formatting for user interaction. **[1 Mark]**

• Web Server: Host software (Apache, Nginx) that listens for incoming HTTP requests, handles SSL certificates, and delivers web resources. **[1 Mark]**

• Scripting Engine: Server-side execution environment (PHP, Python, Node.js) that runs dynamic business logic and processes form inputs. **[1 Mark]**

• Database Backend: Relational database management system (MySQL, PostgreSQL) that securely stores structured user data, product catalogs, and transaction records. **[1 Mark]**